

Reconstructing 360-Degree Memories in Virtual Reality With Generative Artificial Intelligence

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ABSTRACT

Reminiscence is the activity of recalling and revisiting a memory. Past research highlights that reminiscence in older adults can promote healthy aging, improve psychological well-being, and could foster inter-generational communication. Virtual reality offers benefits to reminiscence as it allows users to reenact the past in embodied, immersive, and multi-sensory ways. A key challenge in VR reminiscence is the recreation of past scenes that are relevant and familiar to each user. Thus, this research seeks to evaluate and implement GenAI to prototype 360-degree scenes based on inputs that are relevant to each user. In particular, we examine the implementation 1) text-to-image and 2) panorama outpainting generation methods for 360-degree scenes. This study aims to bring personalization and control to the recreation of memories in VR. Moreover, this work strives to support the democratization of content creation in VR for users with various degrees of technical expertise.

Index Terms: Virtual reality, generative artificial intelligence, older adults.

1 INTRODUCTION

Reminiscence can be described as the activity of recalling and revisiting a memory [11]. In recent years, the exploration of technology-mediated memory recall has sparked growing interest within the field of Human-Computer Interaction [16]. Previous studies have shown that reminiscence can predict the perceived ability to enjoy life [4]. Moreover, past research emphasizes how reminiscence in older adults can promote healthy aging and improve psychological well-being [2]. Thus, the central hypothesis of this study is that the recreation of familiar and past scenes in virtual reality (VR) might improve the user experience in VR and overall well-being in older adults.

Virtual reality might advance reminiscence as it allows users to reenact the past in an embodied, immersive, multi-sensory manner [8]. In particular, older adults could benefit from reminiscence in VR as a tool for inter-generational communication [10]. Reminiscence in VR has been explored from different perspectives. For example, as reminiscence therapy (RT), where users discuss past activities, events, and experiences through familiar items from the past [5]. Similarly, previous research has explored social reminiscence, which involves having several users share a virtual environment and recollect the past through the virtual environment itself [7] or through storytelling [1, 11].

A key challenge in reminiscence is the recreation of past scenes that are relevant and familiar to each user [5]. Since traditional methods for VR prototyping rely on 3D assets to represent virtual environments, it is often challenging to recreate immersive scenes

quickly and inexpensively [15]. Despite the growing interest in reminiscence in VR, to our best knowledge, there have been no attempts to use generative artificial intelligence (GenAI) to recreate past scenes. Thus, this research seeks to evaluate and implement a quick method to prototype immersive scenes based on particular memories that are relevant to each user using GenAI. Our objective is to take advantage of the latest advances in 360-degree image generation so that users can have access to reconstructed scenes that are meaningful to each individual.

2 APPLICATIONS OF REMINISCENCE VR AND GENAI

In this paper, we propose researching GenAI in the context of reminiscence for older adults. We argue that GenAI can advance the creation of reminiscence in VR as it might allow users to recreate personalized immersive scenes [15]. The concept we present relies on visualization and reconstruction of scenes of the past based on the users' input. This approach enables new and previously unimaginable tools to experience and share immersive memories. Our approach includes 1) input from users on the memory to recreate, 2) recreation of the memory as one 360-degree scene given a single input, and 3) interaction methods with GenAI for the manipulation of 360-degree scenes.

2.1 Reminiscence and VR

Previous research has systematically examined the use of immersive VR for reminiscence in older adults, adults 65 years of age or older [9]. Although reminiscence has been associated with the treatment of dementia and other cognitive conditions, research suggests that reminiscence is beneficial for older adults in general [1]. As a result, previous studies focused specifically on older adults with cognitive impairments such as dementia, while others addressed the broader population of older adults.

Several methods have been implemented to recreate immersive reminiscence for older adults. Chapoulie et al. (2014) explored the use of image based rendering (IBR) to recreate familiar and personalized environments [5]. They argue that IBR offers a cost-effective approach to generating realistic 3D models, as it requires only a limited number of input photographs. More recently, there was an attempt to capture the present-day environment using film footage to reconstruct the digital representation of the environment using large-scale 3D Gaussian plating [7]. Meanwhile, others have used more traditional prototyping methods to create VR environment for reminiscence using 3D assets and pre-processing using 3-D engines such as Unity¹ or Unreal². Given the challenges of recreating personalized scenes using conventional VR prototyping techniques, many approaches have turned to storytelling as a way to evoke reminiscence [1, 11].

The growing body of research on immersive VR for reminiscence highlights its potential to enhance well-being among older adults. Although prototyping methods such as Image-Based Rendering (IBR), 3D Gaussian Splatting, and traditional game engine

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¹<https://unity.com/products/unity-engine>

²<https://www.unrealengine.com/en-US>

pipelines have demonstrated the feasibility of recreating immersive environments, they often rely on resource-intensive pre-processing and technical expertise. This dependency not only limits the scalability of VR reminiscence, but also restricts the active participation of older adults in co-authoring and preserving their own memories. There is a critical need for user-friendly tools that empower individuals to create and engage with immersive reminiscence experiences without specialized skills or resources. Bridging this gap will be essential for democratizing VR reminiscence, ensuring that its benefits and use can be widespread. This shift could foster greater autonomy, emotional connection to VR reminiscence experiences.

2.2 GenAI 360-degree scenes

In recent years, several studies have explored 360-degree scene generation using GenAI [15, 12, 15]. There are different approaches that require various input methods and information from the user to generate the 360-image. We have observed and analyzed mainly two types of methods: 1) text-to-image and 2) panorama outpainting generation methods.

The 1) text-to-360-degree approach requires users to input text to generate a 360-degree scenes. For example, PanFusion is a text-to-360-panorama image generation method that can generate panorama images from a single text prompt [15]. However, PanFusion comes with high computational complexity and sometimes fails to generate indoor scenes. Similarly, others have proposed a text-to-3D 360-degree scene generation pipeline that facilitates the creation of 360-degree scenes for in-the-wild environments [17]. Their approach uses the generative power of a 2D diffusion model and prompt self-refinement to create a high-quality, globally coherent panoramic image.

On the other hand, 2) panorama outpainting approaches focus on generating a 360-degree panorama from a partial input image [15]. For example, [6] proposed a training-free pipeline for generating spherical scenes from a single image using multi-modal large language models. However, it seems that current training-free panorama outpainting approaches are still work-in-progress or offer limited tooling for reproducibility [12]. Meanwhile, pre-trained models offer limited generalization [15].

The field of 360-degree scene generation through GenAI is rapidly advancing, with two primary approaches: text-to-image and panorama outpainting. Future work in 360-degree image generation holds promise for bridging gaps in democratizing immersive content creation. This could ultimately facilitate more customization and personalization of immersive experiences.

2.3 Interaction methods

There are limited examples of interaction methods to manipulate the generation of 360-degree scenes using GenAI. Recently, there have been proposals to use voice commands to interact with 360-degree scene generative models [14]. In their approach, users first get to use the system through text-to-2D-degree video. Once the model generates a 2D video, there is backend processing to convert it into 360-degree format. This pre-processing seems conventional; however, once the model has been transformed into 360-degrees, users can use voice commands to generate more videos and adjustments in the 360-degree content. This example highlights potential novel methods for interaction with 360-degree GenAI models.

3 EXPLORATION OF GENAI FOR REMINISCENCE

Our work currently explores the implementation of two GenAI prototyping methods: 1) text-to-360-degree scenes, and 2) photograph-to-360-degree scenes.

First, we believe the use of text-to-360-degree scenes could empower older adults to recreate scenes from memories of places they remember but do not have photographs of, thus offering a different perspective on memory recreation. GenAI prototyping meth-

ods are highly effective in creating static images from text prompts [15, 12, 15]. Thus, our initial work focuses on text-to-360 degree prototyping methods to allow users to generate immersive scenes based on familiar interaction and input methods. Moreover, a text-to-360-degree scene could allow users to visualize and edit the outcome easily by editing the text prompt.

On the other hand, 360-degree prototyping methods through photograph input, a.k.a. panorama outpainting, might enable older adults to use their own photographs as input to recreate memories as 360-degree scenes. This offers great opportunities; however, current panorama outpainting methods are still in development or rely on pre-trained models [6, 12]. This might limit their implementation for our use case. Thus, we are currently examining how to recreate 360-degree scenes based on various, and not single, image inputs. For example, leveraging visual lifelogging images [3] as input for the generation of 360-degree scenes.

4 OPPORTUNITIES AND CHALLENGES OF GENAI FOR REMINISCENCE

Current 360-degree scene prototyping methods are based on the recreation of 3D environments through heavy pre-processing [15]. However, most end users lack the technical expertise or resources required to implement or reproduce these approaches on their own. This represents a major limitation to the democratization of reminiscence content creation in VR. In contrast, GenAI can offer users the opportunity to participate in the recreation of their own memories; allowing users to control/define the input used to recreate a memory. Recreating scenes based on personal scenes of interest could bring more personalized and relevant approaches. For example, recreating the house of infancy would require heavy pre-processing using the current means of VR prototyping but it could be more easily achieved using GenAI prototyping methods. In addition, implementing GenAI prototyping methods for 360-degree scene generation could enable older adults to tell their story from their own perspective to family and friends. This is highly relevant, as previous research suggests that older adults view virtual worlds as useful for inter-generational communication [10].

A key concern with this technology is its psychological impact on users. There is the possibility that the line between the real and synthetic gets blurred, since the generated 360-degree scene will be a mix of real and synthetic elements, from pre-trained references or text/photographic inputs. Users may feel their memories have been altered or mistakenly believe that synthetic elements were part of their actual experiences, rather than recognizing them as additions. Users have been reported to have experienced unpleasant memories evoked by VR [13] and therefore it is critical to examine the quality of the generated output to ensure that users feel psychologically safe to relive memories with the help of GenAI and VR.

Therefore, this research faces technical and ethical challenges moving forward.

5 CONCLUSIONS AND FUTURE WORK

This position paper outlines opportunities and challenges to develop a reminiscence VR system for older adults enriched with generative AI generation of 360-degree scenes. We described the need for personalized reminiscence experiences, previous relevant research, and existing generative AI methods to recreate 360-degree scenes. Our assumption is that combining VR and GenAI for reminiscence in older adults adds a new dimension to the recreation of memories as well as the personalization of memory recreation. Thus, it could set a new milestone in immersive reminiscence. In the near future, this study aims to implement GenAI reminiscence tools that are adequate and adaptable to the needs of older adults. Moreover, we also aspire for this work to support the democratization of VR creation through GenAI, making the process more inclusive and widely available.

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